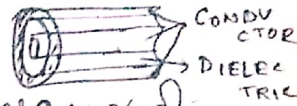


TRANSMISSION LINES

Free space wave propagation is an example of wave propagation in unbounded media which is normally used in radio or TV broadcasting.

Another means of transmitting power or information is by guided structure eg: transmission line or wave guided (coaxial cables, twisted pair, 2 wire line etc).



Transmission line problems are solved by means of EM field theory and circuit theory. But it is easier to solve by circuit theory & deduce the field parameters.

TRANSMISSION LINE PARAMETERS :-

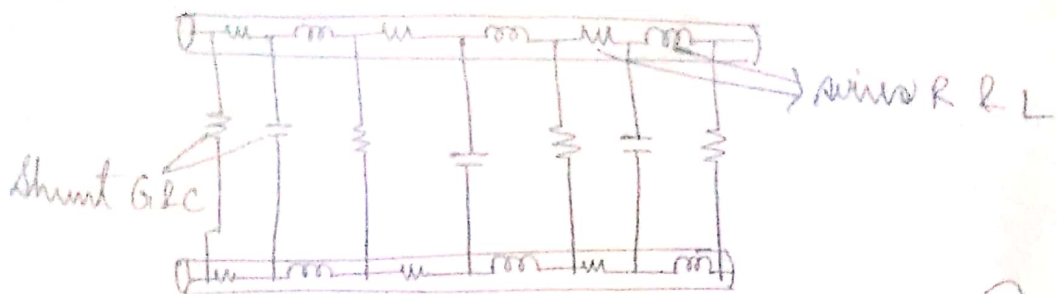
Transmission line can be described by R, L, C & G which are resistance per unit length, inductance per unit length, capacitance per unit length and conductance per unit length.

1. R, L, G, C are not lumped or discrete but are distributed.

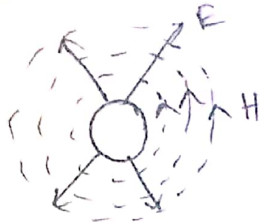
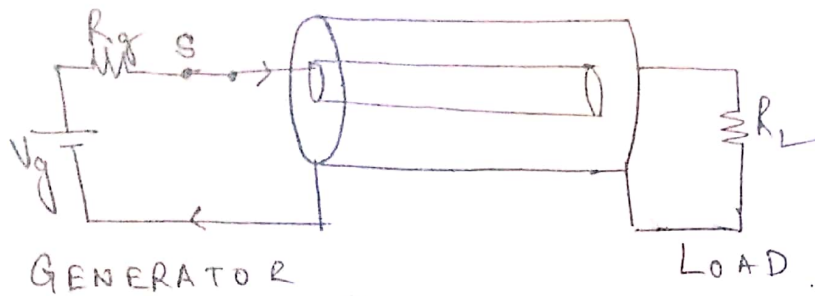
2. For each line the conductors are characterized by $\sigma_c, \epsilon_c = \epsilon_0$ & μ_c & the homogeneous dielectric separating the conductors is characterized by σ, μ, ϵ .

3. $G \neq 1/R$; R : is of conductors comprising the line
 G : of dielectric medium separating the conductors.

4. For each line, $LC = \mu\epsilon$ & $\frac{G}{C} = \frac{\sigma}{\epsilon}$.



Distributed parameters of a two-conductor transmission line



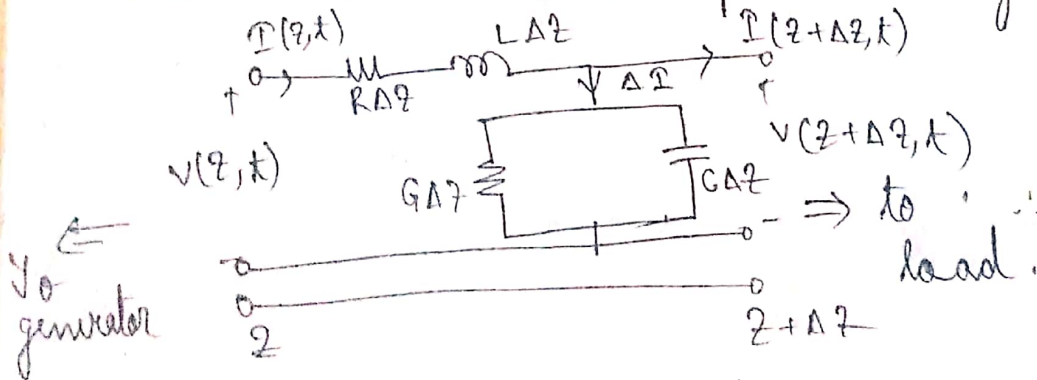
when switch S is closed
the inner conductor is
made +ve w.r.t the outer
one. Then $\vec{E}$ is radially
outward & $\vec{H}$ circles
the current carrying cond.

TRANSMISSION LINE EQUATION

We know,

$$V = - \oint \vec{E} \cdot d\vec{l} \quad \text{and} \quad \oint \vec{H} \cdot d\vec{l} = I \quad \text{--- (1)}$$

Let us consider a small portion Δz of transmission line



Applying KVL in the circuit,

$$V(z, t) = R \Delta z I(z, t) + L \Delta z \frac{\partial I(z, t)}{\partial t} + V(z + \Delta z, t)$$

$$V(z, t) - V(z + \Delta z, t) = R \Delta z I(z, t) + L \Delta z \frac{\partial I(z, t)}{\partial t}$$

$$-\left[\frac{V(z + \Delta z, t) - V(z, t)}{\Delta z} \right] = R I(z, t) + L \frac{\partial I(z, t)}{\partial t}$$

$\Delta z \rightarrow 0$

$$\boxed{-\frac{\partial V(z, t)}{\partial z} = R I(z, t) + L \frac{\partial I(z, t)}{\partial t}} \quad \text{--- (2)}$$

Applying KCL,

$$I(z, t) = I(z + \Delta z, t) + \Delta I$$

$$I(z, t) - I(z + \Delta z, t) = \Delta I$$

$$\text{or } I(z, t) - I(z + \Delta z, t) = G \Delta z V(z + \Delta z, t) + C \Delta z \frac{\partial V(z + \Delta z, t)}{\partial t}$$

$$\frac{I(z, t) - I(z + \Delta z, t)}{\Delta z} = G V(z + \Delta z, t) + C \frac{\partial V(z + \Delta z, t)}{\partial t}$$

or as $\Delta z \rightarrow 0$

$$\boxed{-\frac{\partial I(z, t)}{\partial z} = G V(z, t) + C \frac{\partial V(z, t)}{\partial t}} \quad \text{--- (3)}$$

Taking phasor concept,

$$V(z, t) = \text{Real} (V_s(z) e^{j\omega t}) \quad \text{--- (4)}$$

$$I(z, t) = \text{Real} (I_s(z) e^{j\omega t}) \quad \text{--- (5)}$$

$V_s(z) \Rightarrow$ phasor value of $V(z, t)$
 $I_s(z) \Rightarrow$ " " " $I(z, t)$

Using ④ in ② we get,

$$\begin{aligned} -\frac{dV_s(z)}{dz} &= R I_s(z) + L j\omega I_s(z) \\ &= (R + j\omega L) I_s(z) \end{aligned}$$

$$\therefore \boxed{-\frac{dV_s(z)}{dz} = (R + j\omega L) I_s(z)} \quad \text{--- ⑥}$$

Using ⑤ in ③ we get,

$$\begin{aligned} -\frac{dI_s(z)}{dz} &= G V_s(z) + C j\omega V_s(z) \\ &= (G + j\omega C) V_s(z) \end{aligned}$$

$$\boxed{-\frac{dI_s(z)}{dz} = (G + j\omega C) V_s(z)} \quad \text{--- ⑦}$$

Using ⑦ in ⑥ by differentiating again,

$$+\frac{d^2 V_s(z)}{dz^2} = (R + j\omega L) (G + j\omega C) V_s(z)$$

$$\text{or } \boxed{\frac{d^2 V_s}{dz^2} = (R + j\omega L) (G + j\omega C) V_s} \quad \text{--- ⑧}$$

Differentiating eqn ⑦ & plotting the value of ⑥

$$-\frac{d^2 I_s(z)}{dz^2} = (G + j\omega C) \frac{dV_s(z)}{dz}$$

$$\frac{d^2 I_s(z)}{dz^2} = (G + j\omega C) (R + j\omega L) I_s(z)$$

$$\frac{d^2 I_s}{dz^2} = (G + j\omega C)(R + j\omega L) I_s \quad - (9)$$

let $\gamma^2 = (R + j\omega L)(G + j\omega C)$ & $\gamma = \alpha + j\beta$.

$$\therefore \frac{d^2 I_s}{dz^2} = \gamma^2 I_s \quad - (10)$$

$$\frac{d^2 V_s}{dz^2} = \gamma^2 V_s$$

wavelength, $\lambda = \frac{2\pi}{\beta}$ velocity $u = \frac{\omega}{\beta} = f\lambda$.

Solving equation (10) we get,

$$I_s(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

$$V_s(z) = \underbrace{V_0^+ e^{-\gamma z}}_{+z} + \underbrace{V_0^- e^{\gamma z}}_{-z}$$

V_0^+ & V_0^- are amplitudes of wave travelling in $+z$ & $-z$ direction resp.

Similarly I_0^+ & I_0^- are $+z$ & $-z$ direction wave amplitudes.

$\therefore$ Instantaneous expression for voltage & current,

$$V(z, t) = \text{Re} [V_s(z) e^{j\omega t}] = \text{Re} [(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}) e^{j\omega t}]$$

$$= V_0^+ e^{-\alpha z} \cos(\omega t - \beta z) + V_0^- e^{\alpha z} \cos(\omega t + \beta z) \quad - (11)$$

$\because (\gamma = \alpha + j\beta)$

$$I(z, t) = I_0^+ e^{-\alpha z} \cos(\omega t - \beta z) + I_0^- e^{\alpha z} \cos(\omega t + \beta z) \quad - (12)$$

$$V_0^+ e^{-\gamma z} = (R + j\omega L) I_0^+ e^{-\gamma z}$$

$$\frac{V_0^+}{I_0^+} = \frac{R + j\omega L}{\gamma} = \frac{R + j\omega L}{\sqrt{(R + j\omega L)(G + j\omega C)}}$$

$$\boxed{\frac{V_0^+}{I_0^+} = Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}} \quad \text{--- (17)}$$

Considering -ve travelling wave,

$$-\frac{d}{dz} [V_0^- e^{\gamma z}] = (R + j\omega L) (I_0^- e^{\gamma z})$$

$$-\gamma V_0^- e^{\gamma z} = (R + j\omega L) (I_0^- e^{\gamma z})$$

$$-\frac{V_0^-}{I_0^-} = \frac{R + j\omega L}{\gamma}$$

$$\therefore \boxed{Z_0 = -\frac{V_0^-}{I_0^-} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}}$$

CASE-1 :-

Lossless line ($R=0=G$)

A transmission line is said to be lossless if the conductors of the line are perfect ($\sigma_c \approx \infty$) & the dielectric medium separating them is lossless ($\sigma=0$). Hence $R=0$ & $G=0$.

We know, $\gamma^2 = (R + j\omega L)(G + j\omega C) = (j\omega L)(j\omega C)$

$$\gamma^2 = j^2 \omega^2 LC$$

$$\gamma = j\omega\sqrt{LC} \quad \text{--- (1)}$$

$$\text{Again } \gamma = \alpha + j\beta \quad \text{--- (2)}$$

Comparing (1) & (2) we get,
 $\alpha = 0$ & $\beta = \omega\sqrt{LC}$.

$$\text{Velocity of wave, } u = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \frac{1}{\sqrt{LC}}.$$

$$\text{Characteristic Impedance } Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$\therefore Z_0 = \sqrt{\frac{j\omega L}{j\omega C}} \quad \text{as } (R=0 \text{ \& } G=0)$$

$$\therefore \boxed{Z_0 = \sqrt{L/C}}$$

$$\text{Again } Z_0 = R_0 + jX_0$$

$$\therefore \boxed{R_0 = \sqrt{L/C}}$$

CASE 2: —

Distortionless line ($R/L = G/C$)

$$\text{or } \frac{R}{G} = \frac{L}{C}$$

In a distortionless line the attenuation const α is freq independent while the phase constant β is linearly dependent on freq.

$$\text{We know, } \gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$= \sqrt{R(1+j\omega \frac{L}{R})G(1+j\omega \frac{C}{G})}$$

$$= \sqrt{RG(1+j\omega \frac{C}{G})^2}$$

$$= \sqrt{RG} (1+j\omega \frac{C}{G})$$

$$2 \quad \beta = \omega \sqrt{LC}$$

$$= w \sqrt{\frac{R C^2}{G}} = w \sqrt{\frac{L C^2}{C}} = w \sqrt{L C}$$

or function of freq.

$$= \sqrt{\frac{R}{G_1}} \sqrt{\frac{1+j\omega Y_R}{1+j\omega Y/R}} = \sqrt{R/G_1} = \sqrt{L/C} = R_0 + jX_0$$

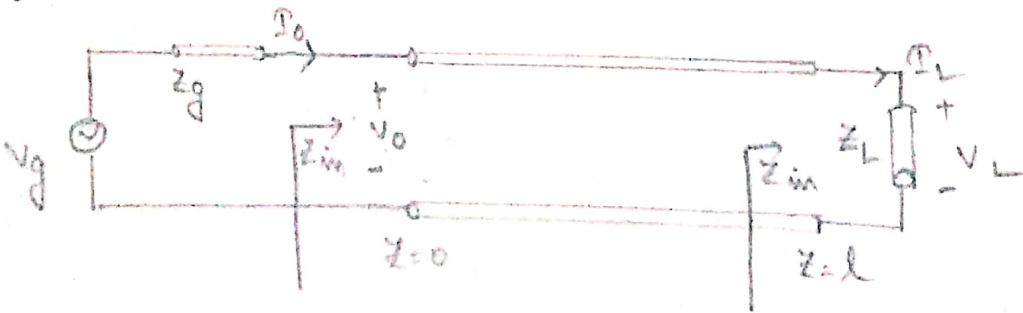
$$R_0 = \sqrt{R/G} = \sqrt{L/C} \quad \& \quad x_0 = 0$$

velocity of wave = $u = \frac{w}{B} = \frac{1}{\sqrt{LC}} = f\lambda$.

A lossless line is a distortionless line but a distortionless line is not necessary lossless.

INPUT IMPEDANCE OF TRANSMISSION LINE

Let us consider a transmission line of length l characterised by propagation constant γ and characteristic impedance Z_0 is connected to load Z_L



Looking into the line, the generator sees the line as an input impedance Z_{in} .



The transmission line extends from $z=0$ at the generator to $z=l$ at the load.

The voltage & current waves can be written as,

$$V_S(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad \text{--- (1)}$$

$$I_S(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

$\xrightarrow{z}$
 $\xleftarrow{z}$

$$\text{or } I_S(z) = \frac{V_0^+}{Z_0} e^{-\gamma z} - \frac{V_0^-}{Z_0} e^{\gamma z} \quad \text{--- (2)} \quad \left[\because Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} \right]$$

At generator side $z=0$ & voltage is V_0 & current is I_0 .

Hence equation (1) can be written as,

$$V_S(0) = V_0 = V_0^+ + V_0^- \quad \text{--- (3)}$$

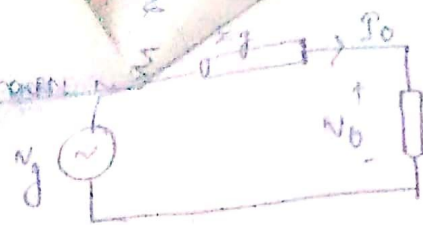
$$I_S(0) = I_0 = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0} \quad \text{--- (4)} \quad (\text{from (2)})$$

$$\textcircled{3} + \textcircled{4}, \quad V_0 + Z_0 I_0 = 2V_0^+$$

$$\therefore \boxed{V_0^+ = \frac{1}{2} (V_0 + Z_0 I_0)} \quad \text{--- (5)}$$

$$\textcircled{3} - \textcircled{4}, \quad V_0 - Z_0 I_0 = 2V_0^-$$

$$\therefore \boxed{V_0^- = \frac{1}{2} (V_0 - Z_0 I_0)} \quad \text{--- (6)}$$



If the input impedance at the input terminals is Z_{in}

the input voltage V_0 , input current I_0 then,

$$V_0 = \left[\frac{Z_{in}}{Z_{in} + Z_g} \right] V_g$$

$$I_0 = \frac{V_g}{Z_{in} + Z_g} \quad \text{--- (7)}$$

V_g : generator voltage

Z_g : generator impedance.

At load side,
 $Z = L$,

voltage at load = V_L
current " " = I_L

We get current and voltage equations as,

$$V_s(x=L) = V_L = V_0^+ e^{-\gamma L} + V_0^- e^{+\gamma L} \quad \text{--- (8)}$$

$$I_s(x=L) = I_L = I_0^+ e^{-\gamma L} + I_0^- e^{+\gamma L}$$

$$= \frac{V_0^+}{Z_0} e^{-\gamma L} - \frac{V_0^-}{Z_0} e^{+\gamma L} \quad \text{--- (9)}$$

$$\therefore \boxed{\begin{aligned} V_L &= V_0^+ e^{-\gamma L} + V_0^- e^{+\gamma L} \quad \text{--- (8)} \\ I_L &= \frac{V_0^+}{Z_0} e^{-\gamma L} - \frac{V_0^-}{Z_0} e^{+\gamma L} \quad \text{--- (9)} \end{aligned}}$$

Solving (8) & (9) we get,

$$V_L + Z_0 I_L = V_0^+ e^{-\gamma L} + V_0^+ e^{-\gamma L}$$

$$V_0^+ = \frac{1}{2} [V_L + Z_0 I_L] e^{+\gamma L} \quad \text{--- (10)}$$

$$V_L - Z_0 I_L = 2V_0^- e^{\gamma l}$$

$$V_0^- = \frac{1}{2}(V_L - Z_0 I_L) e^{-\gamma l}$$

sinh γl
cosh γl

At any point on the line the input impedance

$$Z_{in} = \frac{V_s(z)}{I_s(z)} = \frac{Z_0 (V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z})}{V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}} \quad (\text{from (1) \& (2)})$$

At generator side,

$$Z_{in} = \frac{Z_0 (V_0^+ + V_0^-)}{V_0^+ - V_0^-}$$

$$= Z_0 \left[\frac{\frac{1}{2}(V_L + Z_0 I_L) e^{\gamma l} + \frac{1}{2}(V_L - Z_0 I_L) e^{-\gamma l}}{\frac{1}{2}(V_L + Z_0 I_L) e^{\gamma l} - \frac{1}{2}(V_L - Z_0 I_L) e^{-\gamma l}} \right]$$

$$= Z_0 \left[\frac{(V_L + Z_0 I_L) e^{\gamma l} + (V_L - Z_0 I_L) e^{-\gamma l}}{(V_L + Z_0 I_L) e^{\gamma l} - (V_L - Z_0 I_L) e^{-\gamma l}} \right]$$

$$= Z_0 \left[\frac{I_L (V_L/I_L + Z_0) e^{\gamma l} + Z_0 I_L (V_L/I_L - Z_0) e^{-\gamma l}}{I_L (V_L/I_L + Z_0) e^{\gamma l} - I_L (V_L/I_L - Z_0) e^{-\gamma l}} \right]$$

$$= Z_0 \left[\frac{(Z_L + Z_0) e^{\gamma l} + (Z_L - Z_0) e^{-\gamma l}}{(Z_L + Z_0) e^{\gamma l} - (Z_L - Z_0) e^{-\gamma l}} \right] \quad \text{--- (11)}$$

Now, $\sinh \gamma l = \frac{e^{\gamma l} - e^{-\gamma l}}{2}$ $\cosh \gamma l = \frac{e^{\gamma l} + e^{-\gamma l}}{2}$

$$2 \tanh \gamma l = \frac{e^{\gamma l} - e^{-\gamma l}}{e^{\gamma l} + e^{-\gamma l}} \quad \text{--- (12)}$$

distance

$$\left. \begin{aligned} \sinh \gamma l + \cosh \gamma l &= \frac{2e^{\gamma l}}{2} = e^{\gamma l} \\ \cosh \gamma l - \sinh \gamma l &= e^{-\gamma l} \end{aligned} \right\} \text{--- (13)}$$

Now Putting the values of 13 in equation (11),

$$Z_{in} = Z_0 \left[\frac{(Z_L + Z_0)(\sinh \gamma l + \cosh \gamma l) + (Z_L - Z_0)(\cosh \gamma l - \sinh \gamma l)}{(Z_L + Z_0)(\sinh \gamma l + \cosh \gamma l) - (Z_L - Z_0)(\cosh \gamma l - \sinh \gamma l)} \right]$$

$$Z_{in} = Z_0 \left[\frac{Z_L \sinh \gamma l + Z_0 \sinh \gamma l + Z_L \cosh \gamma l + Z_0 \cosh \gamma l + Z_L \cosh \gamma l - Z_0 \sinh \gamma l - Z_L \sinh \gamma l + Z_0 \sinh \gamma l}{Z_L \sinh \gamma l + Z_0 \sinh \gamma l + Z_L \cosh \gamma l + Z_0 \cosh \gamma l - Z_L \cosh \gamma l + Z_0 \cosh \gamma l - Z_0 \sinh \gamma l + Z_L \sinh \gamma l} \right]$$

$$= Z_0 \left[\frac{2 Z_0 \sinh \gamma l + 2 Z_L \cosh \gamma l}{2 Z_L \sinh \gamma l + 2 Z_0 \cosh \gamma l} \right]$$

$$\begin{aligned} Z_{in} &= Z_0 \left[\frac{Z_0 (e^{\gamma l} - e^{-\gamma l}) + Z_L (e^{\gamma l} + e^{-\gamma l})}{Z_0 (e^{\gamma l} + e^{-\gamma l}) + Z_L (e^{\gamma l} - e^{-\gamma l})} \right] \\ &= Z_0 \left[\frac{Z_0 \sinh \gamma l + Z_L \cosh \gamma l}{Z_0 \cosh \gamma l + Z_L \sinh \gamma l} \right] \end{aligned}$$

$$Z_{in} = Z_0 \left[\frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l} \right] \text{--- (14)}$$

For a lossless line $\gamma = \alpha + j\beta = j\beta$ as $\alpha = 0$.

$$\therefore Z_{in} = Z_0 \left[\frac{Z_L + Z_0 \tanh j\beta l}{Z_0 + Z_L \tanh j\beta l} \right]$$

$$= Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right] \text{--- (15)}$$

For any transmission line,

$$\text{Input Impedance } Z_{in} = Z_0 \left[\frac{Z_L + Z_0 \tanh \beta l}{Z_0 + Z_L \tanh \beta l} \right] \quad (\text{lossy line})$$

$\therefore Z_{in}$ varies periodically.

$$Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right] \quad \text{for lossless}$$

LOSSY LINE :-

For short circuited i.e. load $Z_L = 0$,

$$Z_{in} = Z_0 \left[\frac{Z_0 \tanh \beta l}{Z_0} \right] = Z_0 \tanh \beta l$$

For open circuited, $Z_L = \infty$

$$Z_{in} = Z_0 \coth \beta l$$

For matched line, $Z_L = Z_0$

$$Z_{in} = Z_0$$

(considering $\alpha = 0$)

$$\begin{aligned} Z_{in} &= \frac{Z_0 (Z_L + Z_0 \tanh \beta l)}{(Z_0 + Z_L \tanh \beta l)} \\ &= Z_0 \left[\frac{1 + \frac{Z_0}{Z_L} \tanh \beta l}{\frac{Z_0}{Z_L} + \tanh \beta l} \right] \\ &= Z_0 \left[\frac{1 + 0}{0 + \tanh \beta l} \right] \\ \therefore Z_{in} &= Z_0 \coth \beta l \end{aligned}$$

The quantity βl is electrical length of the line expressed in degree or radians.

LOSSLESS LINE :-

For short circuited, $Z_{in} = j Z_0 \tan \beta l$ ($Z_L = 0$)

For open circuited, $Z_{in} = \lim_{Z_L \rightarrow \infty} \frac{Z_0}{j \tan \beta l} = -j Z_0 \cot \beta l$ ($Z_L = \infty$)

For matched $Z_{in} = Z_0$

VOLTAGE REFLECTION CO-EFFICIENT :-

Reflection co-efficient at load is the ratio of reflected wave to the incident wave,

$$\Gamma_L = \frac{V_o^- e^{\gamma l}}{V_o^+ e^{-\gamma l}}$$

$$V_s(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

Again at load, $V_o^+ = \frac{1}{2} (V_L + Z_0 I_L) e^{\gamma l}$

$$V_o^- = \frac{1}{2} (V_L - Z_0 I_L) e^{-\gamma l}$$

$$\therefore \Gamma_L = \frac{\frac{1}{2} (V_L - Z_0 I_L) e^{-\gamma l} \cdot e^{\gamma l}}{\frac{1}{2} (V_L + Z_0 I_L) e^{-\gamma l} \cdot e^{\gamma l}}$$

$$= \frac{V_L - Z_0 I_L}{V_L + Z_0 I_L} = \frac{V_L/I_L - Z_0}{V_L/I_L + Z_0}$$

$$\boxed{\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

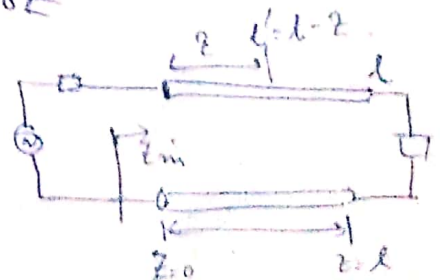
Voltage reflection co-efficient at any point on the line is the ratio of the reflected voltage wave to that of the incident wave.

$$\Gamma(z) = \frac{V_o^- e^{\gamma z}}{V_o^+ e^{-\gamma z}} = \frac{V_o^-}{V_o^+} e^{+2\gamma z}$$

But $z = l - l'$,

$$\Gamma(z) = \frac{V_o^-}{V_o^+} e^{2\gamma l} e^{-2\gamma l'}$$

$$\boxed{\Gamma(z) = \Gamma_L e^{-2\gamma l'}}$$



CURRENT REFLECTION COEFFICIENT:

$$= \frac{\text{reflected current wave}}{\text{incident current wave}} = \frac{I_0^- e^{\gamma L}}{I_0^+ e^{-\gamma L}}$$
$$= -\Gamma_L$$

STANDING WAVE :- When the transmitted ^{signal} wave is not correctly terminated, the travelling EM wave is reflected either completely or partially at the termination. The interference of incident and reflected waves gives standing waves of current & voltage along the line. It is the ratio of maximum to minimum current or voltage on the line,

$$SWR = \text{standing wave ratio} = S = \left| \frac{V_{\max}}{V_{\min}} \right|$$
$$S = \left| \frac{I_{\max}}{I_{\min}} \right|$$

S gives the amt of mismatch between load & transmission line

$S > 1 \Rightarrow$ mismatching (produces noise, power loss)

$S = 1 \Rightarrow$ totally matched

$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

or

$$|\Gamma_L| = \frac{S - 1}{S + 1}$$

we know,

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{V_0^- e^{j\beta l}}{V_0^+ e^{-j\beta l}}$$

$$\& \quad \Gamma(z) = \frac{V_0^-}{V_0^+} e^{2j\beta z}$$

$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \quad \text{or} \quad \Gamma_L = \frac{S - 1}{S + 1}$$

For a lossless line,

$$\Gamma_L = 0, \quad S = 1$$

$$\Gamma_L = -1, \quad S \rightarrow \infty$$

$$\Gamma_L = 1, \quad S \rightarrow \infty$$

when $Z_L = Z_0$ (matched load)

when $Z_L = 0$ (short circuited)

when $Z_L \rightarrow \infty$ (open circuited)

(ii) if Z_L & Z_0 are real,

(i) $Z_L > Z_0$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$S = \frac{1 + \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right|}{1 - \left| \frac{Z_L - Z_0}{Z_L + Z_0} \right|} = \frac{Z_L + Z_0 + Z_L - Z_0}{Z_L + Z_0 - Z_L + Z_0} = \frac{Z_L}{Z_0}$$

$$(ii) \text{ if } Z_L < Z_0, \quad \Gamma_L = \frac{Z_0 - Z_L}{Z_0 + Z_L}$$

$$S = \frac{1 + \frac{Z_0 - Z_L}{Z_0 + Z_L}}{1 - \frac{Z_0 - Z_L}{Z_0 + Z_L}} = \frac{Z_0}{Z_L}$$

POWER TO BE DELIVERED TO LOAD

Voltage at load $V_L(l) = (V_0^+ e^{-j\beta l} + V_0^- e^{j\beta l})$

$$= V_0^+ e^{-j\beta l} + \Gamma_L V_0^+ e^{-j\beta l}$$

$$= V_0^+ e^{-j\beta l} (1 + \Gamma_L) \quad (\text{for lossless } \Gamma_L = \frac{Z_L}{Z_0} = \frac{Z_L}{Z_0})$$

$$I_S(l) = I_0^+ e^{-j\beta l} + I_0^- e^{j\beta l}$$

$$= \frac{V_0^+}{Z_0} e^{-j\beta l} - \frac{V_0^-}{Z_0} e^{j\beta l}$$

$$= \frac{V_0^+}{Z_0} e^{-j\beta l} (1 - \Gamma_L)$$

$$P = \frac{1}{2} \text{Re} [V_S(l) I_S^*(l)]$$

$\rightarrow$ conjugate
 $Z_0 \rightarrow$ real for lossless

$$= \frac{1}{2} \text{Re} \left[\frac{|V_0^+|^2}{Z_0} (1 - |\Gamma_L|^2) \right]$$

$$= \frac{1}{2} \frac{|V_0^+|^2}{Z_0} (1 - |\Gamma_L|^2)$$

$$= P_i - P_r \quad P_i = \frac{1}{2} \frac{|V_0^+|^2}{Z_0}$$

$$P_r = -\frac{1}{2} \frac{|V_0^+|^2}{Z_0} |\Gamma_L|^2$$

$$= P_{\text{incident}} - P_{\text{reflected}}$$

Maximum power will be delivered to load when $\Gamma_L = 0$.

Group velocity $v_g = \frac{d\omega}{d\beta}$

TRANSMISSION LINE AS CIRCUIT ELEMENTS

Transmission line act as wave-guiding structure under normal frequencies. At very high frequencies it can be used as circuit elements like a capacitor or an inductor.

considering the line to be lossless,

$$\gamma = \alpha + j\beta = j\beta$$

and characteristic impedance $Z_0 = R_0 + jX_0 \approx R_0$

$$\therefore \tanh \beta l = \tanh(j\beta l) = j \tan \beta l$$

$$\therefore Z_{in} = Z_0 \left[\frac{Z_L + Z_0 \tanh \beta l}{Z_0 + Z_L \tanh \beta l} \right] = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right]$$

$$= R_0 \left[\frac{Z_L + j R_0 \tan \beta l}{R_0 + j Z_L \tan \beta l} \right]$$

$$\begin{aligned} Z_0 &= \sqrt{L/C} \\ &= \text{Real} \\ &= R_0 \end{aligned}$$

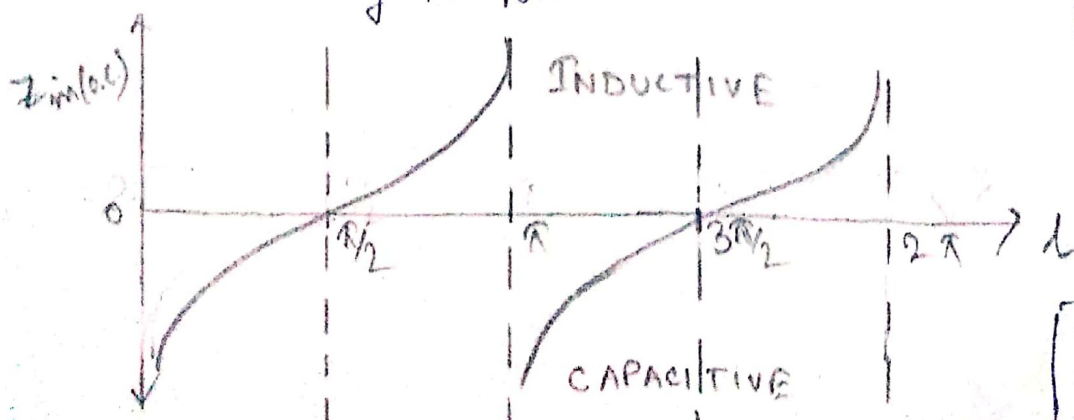
SPECIAL CASE :-

• If open circuited :-

$$Z_L = \infty$$

$$\therefore Z_{in}(o.c) = R_0 \left[\frac{1 + j \frac{R_0}{Z_L} \tan \beta l}{\frac{R_0}{Z_L} + j \tan \beta l} \right]$$

$$= \frac{R_0}{j \tan \beta l} = -j R_0 \cot \beta l$$



$$\begin{aligned} \Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{1 - Z_0/Z_L}{1 + Z_0/Z_L} \\ &= 1 \end{aligned}$$

$$\boxed{\begin{aligned} S &= \infty \\ \Gamma_L &= 1 \end{aligned}}$$

SHORT-CIRCUITED :-

$$Z_L = 0$$

$$Z_{in(sc)} = Z_0 \left[\frac{Z_L + R_0 Z_0 j \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right] = R_0 \left[\frac{Z_L + R_0 j \tan \beta l}{R_0 + j Z_L \tan \beta l} \right]$$
$$= R_0 \left[\frac{0 + R_0 j \tan \beta l}{R_0 + 0} \right] = j R_0 \tan \beta l$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -1$$

$$S = \infty$$

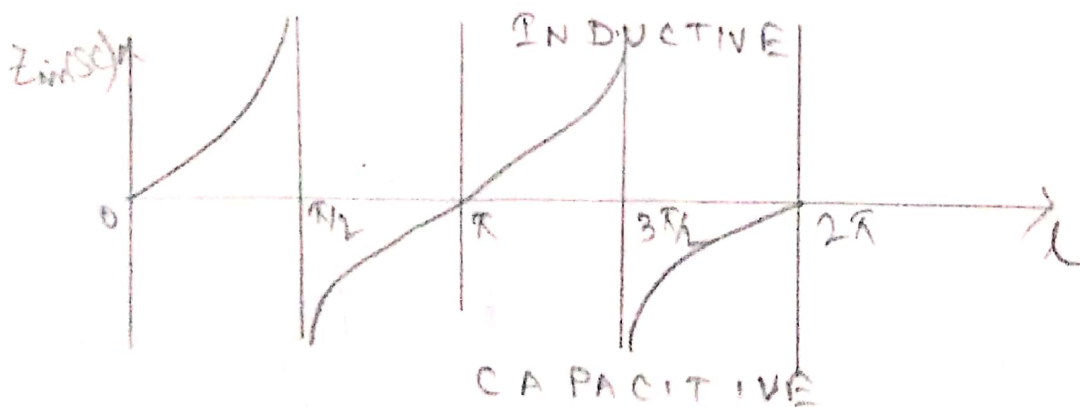
Z_{in} is a pure reactance, which could be capacitive or inductive depending on the value of l .

$\beta l \ll 1$ (short length)

then $\tan \beta l \approx \beta l$.

$$Z_{in(sc)} = j R_0 \beta l = j \left(\sqrt{\frac{L}{C}} \right) (\omega \sqrt{LC}) l$$
$$= j \omega L l$$

$R_0 = \sqrt{L/C}$



$$\begin{aligned} \tan 2\pi &= 0 \\ \tan \pi &= 0 \\ \cot \pi &= -1 \\ \cot \pi/2 &= 0 \end{aligned}$$

$$X_L = \omega L$$

$$X_C = \frac{1}{\omega C}$$

QUARTER-WAVE TRANSMISSION LINE ($l = \lambda/4$)

$$l = \lambda/4$$

$$\therefore \beta l = \left(\frac{2\pi}{\lambda} \right) \cdot \left(\lambda/4 \right) = \pi/2$$

$$Z_{in} = R_0 \left[\frac{Z_L + j R_0 \tan \beta l}{R_0 + j Z_L \tan \beta l} \right]$$

When $l = \text{odd multiples of } \lambda/4 = (2m-1)(\lambda/4)$ with $m=1, 2, 3, \dots$

$$\beta l = \left(\frac{2\pi}{\lambda} \right) \left[(2m-1)(\lambda/4) \right] = (2m-1)(\pi/2)$$

$$\boxed{\tan \beta l = \tan \left[(2m-1)(\pi/2) \right] \rightarrow \pm \infty}$$

$$\therefore Z_{in} = R_0 \left[\frac{\frac{Z_L}{\tan \beta l} + j R_0}{\frac{R_0}{\tan \beta l} + j Z_L} \right] = \frac{R_0^2}{Z_L} \quad (\text{for lossless})$$

$$\text{or } Z_{in} = \frac{Z_0^2}{Z_L} \Rightarrow \frac{Z_{in}}{Z_0} = \frac{Z_0}{Z_L} \Rightarrow Z_{in} = Z_0/Z_L$$

The above equation shows the impedance at the input of a quarter wave line depends on two quantities, load impedance (which is fixed for any load at a constant frequency) and characteristic impedance.

HALF-WAVE TRANSMISSION LINE ($l = \lambda/2$):

When the line length equals half-wave length,
 $l = \lambda/2$.

$$\therefore \beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} = \pi$$

When the line length is an integral multiple of $\lambda/2$,
 $l = m \lambda/2$ with $m = 1, 2, 3, \dots$

$$\beta l = \left(\frac{2\pi}{\lambda}\right) n\left(\frac{\lambda}{2}\right) = n\pi$$

$$\therefore \tan \beta l = \tan(n\pi) = 0$$

$$\therefore Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right] = Z_0 \left[\frac{Z_L}{Z_0} \right]$$

$$\boxed{Z_{in} = Z_L}$$

A half-wave lossless line transforms the load impedance to the input terminals without change.

TRANSMISSION LINE OF LENGTH $\lambda/8$:-

$$l = \lambda/8$$

$$\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{8} = \frac{\pi}{4}$$

$$Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right] = Z_0 \left[\frac{Z_L + j Z_0 \tan(\pi/4)}{Z_0 + j Z_L \tan(\pi/4)} \right]$$

$$= Z_0 \left[\frac{Z_L + j Z_0}{Z_0 + j Z_L} \right]$$

$$\tan(\pi/4) = 1$$

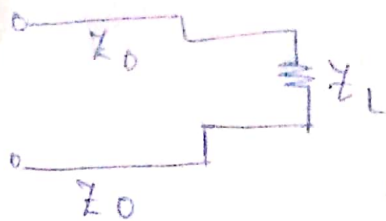
When $Z_L = 0$, $Z_{in} = j Z_0$

When $Z_L = \infty$, $Z_{in} = -j Z_0$

For lossless line impedance is resistive. In that case matching is done by $\lambda/4$ transformer or Transmission line. For complex load stub is used.

IMPEDANCE MATCHING

If $Z_0 = Z_L$ Then $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = 0$



$\therefore$ no reflection for perfectly matched & maximum power is transferred.

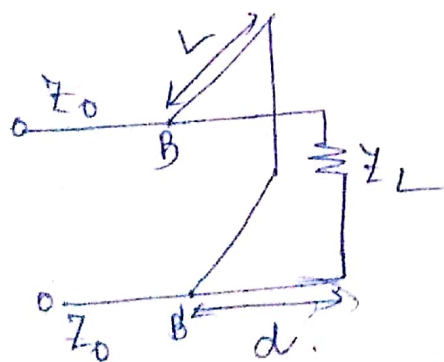
If $Z_0 \neq Z_L$; $\Gamma \neq 0 \Rightarrow$ reflection occurs.

Techniques are available to match Z_0 & Z_L by joining elements serially or parallelly to the line.

(i) Half-wave Transformer ($\lambda/2$)

(ii) Stub:—

Stub:— stub matches are used to match any complex load to a transmission line. ~~the stub~~ stub can be single or double. It can be connected serially or parallelly. It can be open or short circuited.



L : length of stub
 d : distance of " from load.
 stub is connected parallelly to the line & it is short circuited.

$$Z_L = r + jx$$

$$\frac{Z_L}{Z_0} = r' + jx' = \tilde{Z}_L'$$

as " it is better

To consider admittance (because admittance can be added in ")

$$Z_o = \frac{1 + \frac{|V_r|}{|V_f|}}{1 - \frac{|V_r|}{|V_f|}} = Z_o \left[\frac{1 + |\rho|}{1 - |\rho|} \right] \quad \dots(6.86a)$$

From equation (6.83) and (6.86)

$$\boxed{Z_{\max} = Z_o S_v} \quad \dots(6.87)$$

Maximum impedance of a line is the product of the characteristic impedance and voltage standing wave ratio (S_v).

Similarly,

$$Z_{\min} = \frac{V_{\min}}{I_{\max}}$$

$$V_{\min} = |V_f| - |V_r|$$

$$I_{\max} = \frac{|V_f| + |V_r|}{Z_o}$$

Put these values

$$Z_{\min} = \frac{|V_f| - |V_r|}{\frac{|V_f| + |V_r|}{Z_o}} = Z_o \left[\frac{|V_f| - |V_r|}{|V_f| + |V_r|} \right]$$

$$= Z_o \left[\frac{1 - \frac{|V_r|}{|V_f|}}{1 + \frac{|V_r|}{|V_f|}} \right] = Z_o \left[\frac{1 - |\rho|}{1 + |\rho|} \right]$$

$$= Z_o \left[\frac{1}{\frac{1 + |\rho|}{1 - |\rho|}} \right] = Z_o \left[\frac{1}{S_v} \right]$$

$$\therefore \boxed{Z_{\min} = \frac{Z_o}{S_v}} \quad \dots(6.88)$$

Thus, minimum impedance of the line is the product of the characteristic impedance and reciprocal of voltage standing wave ratio (VSWR).

6.11. POWER TRANSMITTED

$$Z_{\max} = \frac{V_{\max}}{I_{\min}}$$

Also from equation (6.87)

$$Z_{\max} = Z_o S$$

Then $Z_o S = \frac{V_{\max}}{I_{\min}}$

$\therefore V_{\max} = I_{\min} Z_o S$

$$I_{\min} = \frac{V_{\max}}{Z_o S}$$

Power transmitted = $P = |V_{\max}| \times |I_{\min}|$

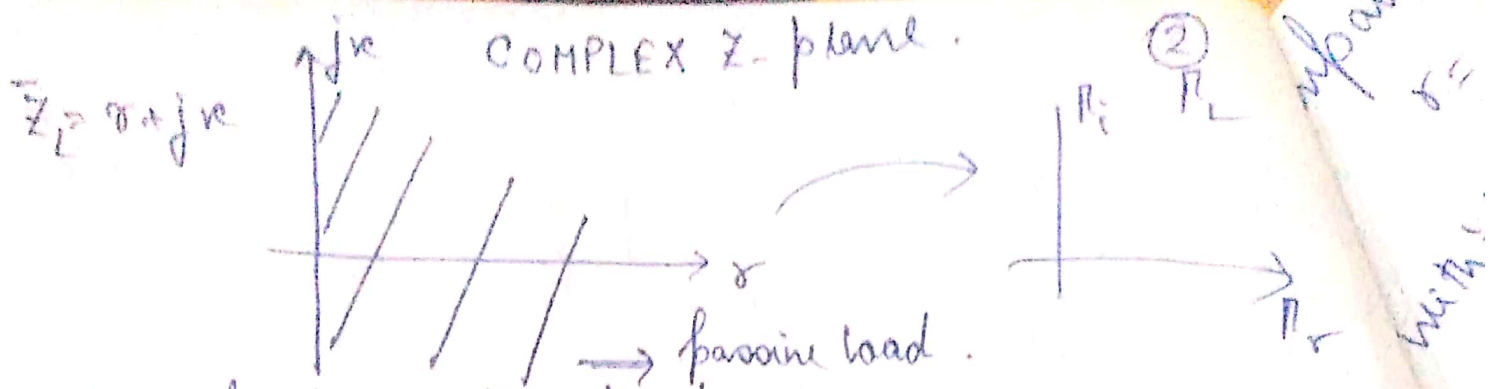
$$= |V_{\max}| \times \frac{|V_{\max}|^2}{Z_o S}$$

$$P = \frac{|V_{\max}|^2}{Z_o S}$$

Thus, power transmitted can be calculated by equation (6.89).

6.12. APPLICATIONS OF TRANSMISSION LINES

1. Transmission lines are used for transmitting the energy and signal from one point to another point.
2. Transmission lines are used for the transmission of electrical energy from the generating stations to the load centres.
3. Transmission lines are used for distributing the power.
4. Two wire transmission line is used in power system or telephone lines.
5. Coaxial transmission line are used as TV cables, telephone cables.
6. Optical fibre are used for transference of loss-less light signals.
7. They are also used to connect the antenna to the TV receiver.



$x=0 \rightarrow$ purely reactive load

For every passive load there is a value of Γ .

$$\Gamma = \Gamma_r + j\Gamma_i = \frac{\bar{Z}_L - 1}{\bar{Z}_L + 1} = \frac{(r+jx) - 1}{(r+jx) + 1} = \text{Ratio in polar}$$

$$\text{or } \bar{Z}_L = \frac{1 + \Gamma}{1 - \Gamma} \quad \text{or } (r+jx) = \frac{1 + (\Gamma_r + j\Gamma_i)}{1 - (\Gamma_r + j\Gamma_i)}$$

Conversion of Impedance $\rightarrow$ reflection coeff.

$$r+jx = \frac{\Gamma_r + j\Gamma_i + 1}{1 - \Gamma_r - j\Gamma_i} = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}$$

$$= \frac{[(1 + \Gamma_r) + j\Gamma_i][(1 - \Gamma_r) + j\Gamma_i]}{[(1 - \Gamma_r) - j\Gamma_i][(1 - \Gamma_r) + j\Gamma_i]}$$

$$= \frac{(1 + \Gamma_r)(1 - \Gamma_r) + j\Gamma_i(1 - \Gamma_r) + j\Gamma_i(1 + \Gamma_r) + j^2\Gamma_i^2}{(1 - \Gamma_r)^2 - (j\Gamma_i)^2}$$

$$= \frac{1 - \Gamma_r^2 - \Gamma_i^2 + j^2\Gamma_i - j\Gamma_i\Gamma_r + j\Gamma_i + j\Gamma_i\Gamma_r}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

$$= \frac{1 - \Gamma_r^2 - \Gamma_i^2 + 2j\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2} = \frac{(1 - \Gamma_r^2) - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} + j \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

unpairing,

$$r = \frac{1 - \pi_r^2 - \pi_i^2}{(1 - \pi_r)^2 + \pi_i^2}$$

$$\textcircled{3} \quad r = \frac{2\pi_i}{(1 - \pi_r)^2 + (\pi_i)^2}$$

with π_i^2 ,

$$r = \frac{1 - \pi_r^2 - \pi_i^2}{(1 - \pi_r)^2 + \pi_i^2}$$

$$r(1 - \pi_r)^2 + r\pi_i^2 = 1 - \pi_r^2 - \pi_i^2$$

$$r(1 - 2\pi_r + \pi_r^2) + r\pi_i^2 = 1 - \pi_r^2 - \pi_i^2$$

$$r - 2r\pi_r + r\pi_r^2 + r\pi_i^2 - 1 + \pi_r^2 + \pi_i^2 = 0$$

$$r\pi_r^2 + \pi_r^2 - 2r\pi_r + r - 1 + \pi_i^2 + r\pi_i^2 = 0$$

$$(\pi_r^2 - 2r\pi_r + (r-1) + \pi_i^2(r+1)) = 0$$

Dividing by $(r+1)$

$$\pi_r^2 - 2\left(\frac{r}{r+1}\right)\pi_r + \left(\frac{r-1}{r+1}\right) + \pi_i^2 = 0$$

Introducing $\left(\frac{r}{r+1}\right)^2$ both sides,

$$\pi_r^2 - 2\left(\frac{r}{r+1}\right)\pi_r + \left(\frac{r}{r+1}\right)^2 + \pi_i^2 + \frac{r-1}{r+1} = \left(\frac{r}{r+1}\right)^2$$

$$\left[\pi_r - \frac{r}{r+1}\right]^2 + \pi_i^2 = \left(\frac{r}{r+1}\right)^2 - \frac{r-1}{r+1}$$

$$= \left(\frac{r}{r+1}\right)^2 - \frac{(r-1)(r+1)}{(r+1)^2}$$

$$= \left(\frac{r}{r+1}\right)^2 - \frac{r^2 - 1}{(r+1)^2}$$

$$\left(\pi_r - \frac{r}{r+1}\right)^2 + \pi_i^2 = \left(\frac{1}{r+1}\right)^2$$

$$\left[\Pi_r - \frac{r}{1+r} \right]^2 + \Pi_i^2 = \left(\frac{1}{1+r} \right)^2$$

This is an equation of circle $(x-h)^2 + (y-k)^2 = a^2$ whose centre is at (h, k) & radius a

centre $\rightarrow \left(\frac{r}{1+r}, 0 \right)$

radius $\rightarrow \frac{1}{1+r}$

Similarly, solving we get,

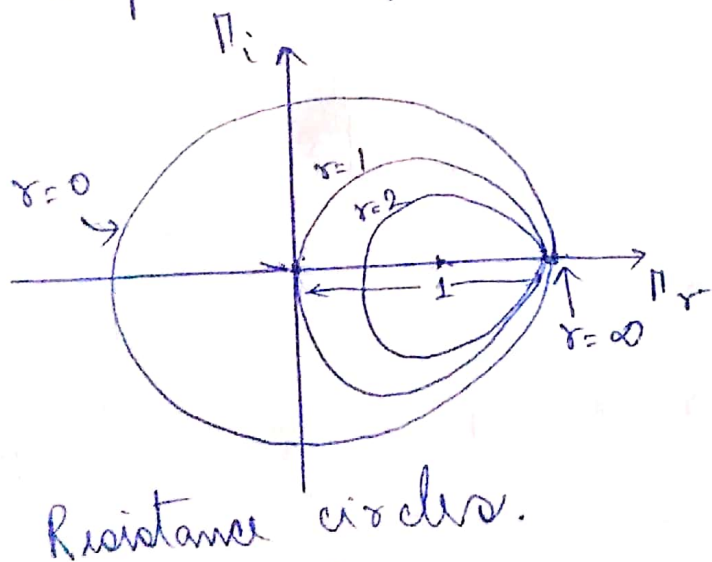
$$(\Pi_r - 1)^2 + \left(\Pi_i - \frac{1}{r} \right)^2 = \left(\frac{1}{r} \right)^2$$

radius is $\frac{1}{r}$

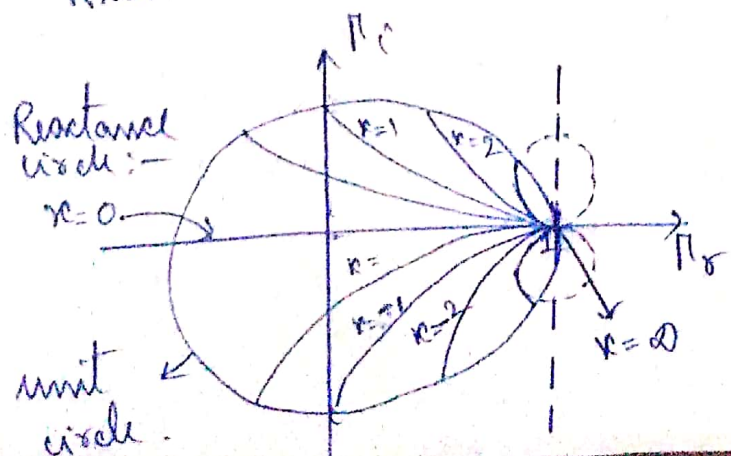
(here Π_r is always 1)

$r \rightarrow +\infty$: Inductive
 $r \rightarrow -\infty$: Capacitive

Complex Π plane:-

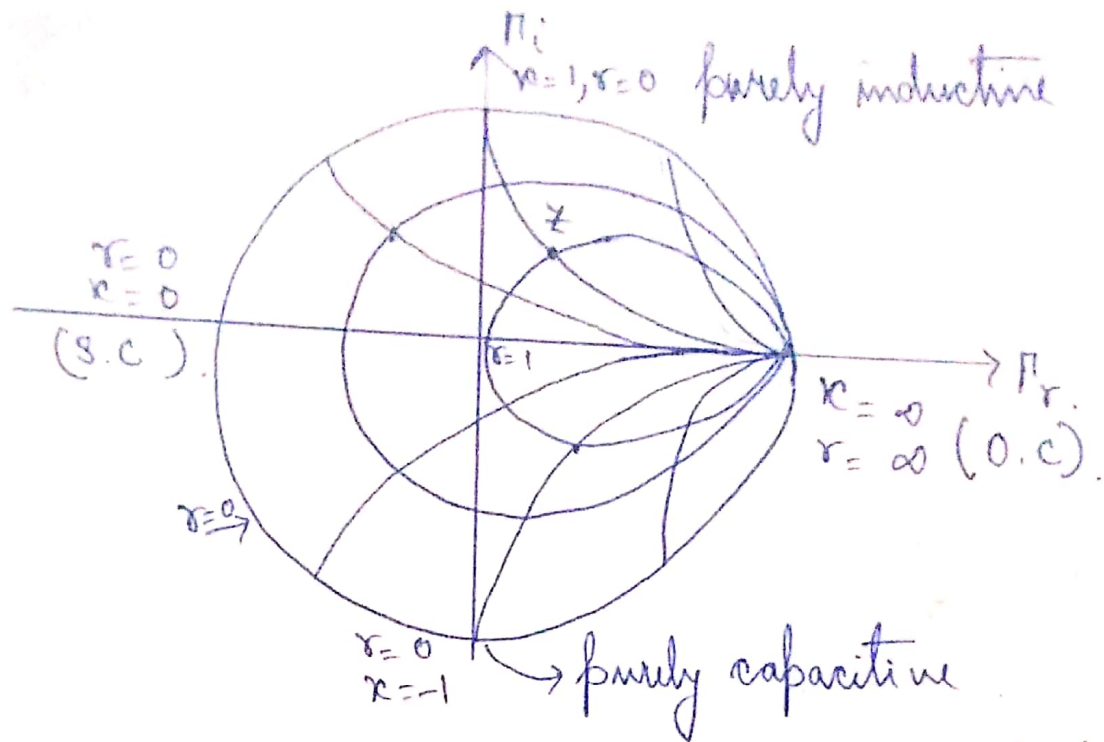


r	RADIUS $\left[\frac{1}{r+1} \right]$	CENTRE $\left[\frac{r}{r+1}, 0 \right]$
0	1	0, 0
1	0.5	0.5, 0
2	$\frac{1}{3}$	$\frac{2}{3}, 0$
$\vdots$		
∞	$\frac{1}{\infty+1} = 0$ (pt)	$\frac{\infty}{\infty+1}, 0$ $= (1, 0)$

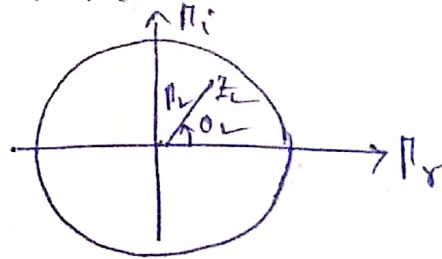


r	RADIUS $\left(\frac{1}{r} \right)$	CENTRE $\left(1, \frac{1}{r} \right)$
0	$\frac{1}{0} = \infty$	$1, \frac{1}{0} \rightarrow$ line
1	$\frac{1}{1} = 1$	1, 1
2	$\frac{1}{2}$	$1, \frac{1}{2}$
3	$\frac{1}{3}$	$1, \frac{1}{3}$
$\vdots$		
∞	0 (pt)	$1, \frac{1}{\infty} = 1, 0$
-1	-1	1, -1

SWR CHART



Reflection coefficient at load :- $\Gamma_L = |\Gamma_L| e^{j\theta_L}$ — (1)



" " at any pt on Transmission line :-

$$\Gamma = \Gamma_L e^{-j2\beta l} = \Gamma_L e^{-2\beta l}$$

$$= \Gamma_L e^{-2j\beta l}$$

$$= |\Gamma_L| e^{j(\theta_L - 2\beta l)}$$

$\beta = \alpha + j\beta$
 $= j\beta$ (lossless)
 l : distance from load

— (2)

$(\theta_L - 2\beta l)$: total phase $\angle$ of ref coeff which changes

with l & mag is $|\Gamma_L|$ which is constant at all pts on TR line or when we move on TR line as we move phase changes

if $l: +ve$, we move towards generator & the phase becomes more $-ve$

movement of on TR_x line is equivalent to the rotation over circle.

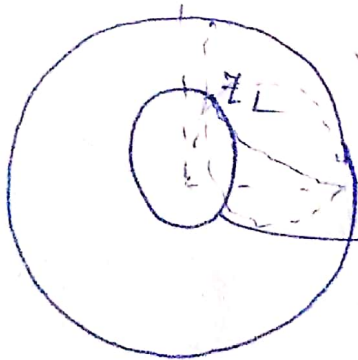
$$2\pi = 2\beta l$$

$$l = \lambda/2$$

$\therefore$ a full rotation over Smith chart is equivalent to length $\lambda/2$.

Three scales are included at the periphery of the Smith chart (1 $\rightarrow$ protractor in degrees, 2 $\rightarrow$ distance on the line from generator & outermost calculates distance on the line from generator in λ). One scale is sufficient. They are used to determine the distance from load or generator in degrees or wavelengths.

a distance $\lambda/2$ corresponds to 360° or 2π .



We know Γ_L is const on the circle drawn with the centre as ~~radius~~ origin as centre for a TR_x line.

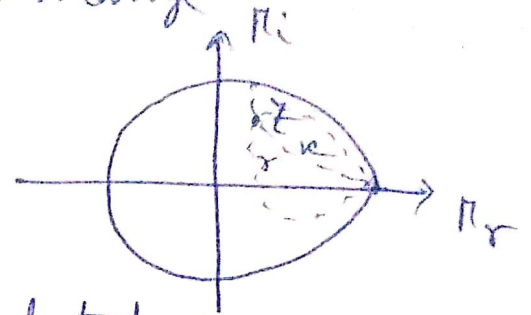
$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = 1$$

If a point corresponding to some Z_L is closer to origin w.r.t another pt then reflection is less as $|\Gamma_L|$ is less. Better match.
 $\Rightarrow$ c.w movement $\rightarrow$ towards generator / a.c.w $\rightarrow$ towards load

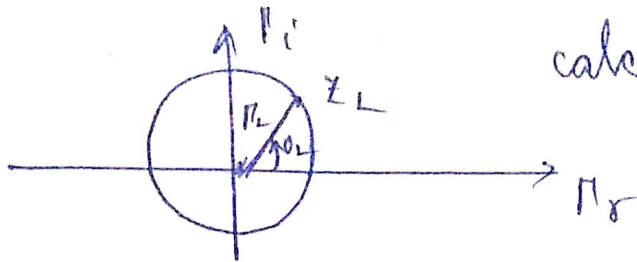
APPLICATION OF SMITH CHART

when Z_L is given. To normalize

$$\bar{Z}_L = \frac{Z_L}{Z_0} = r + jx$$

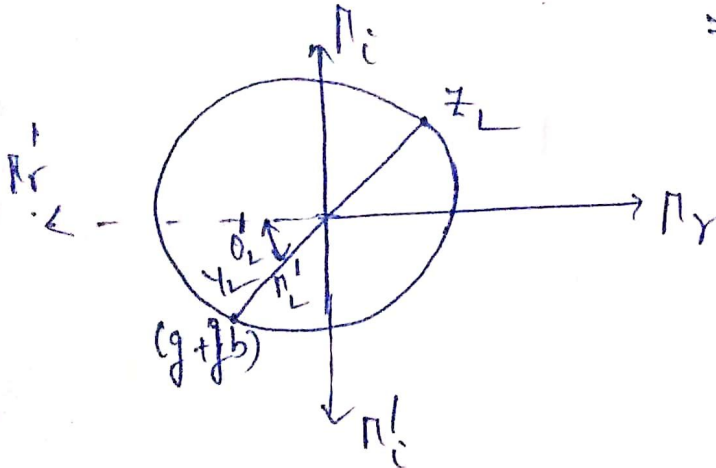


2) Draw VSWR circle



calculate $| \Gamma_L |$ & θ_L .

3) calculate Γ_L .



diametrically opp to z_L on VSWR circle (with center along its path)

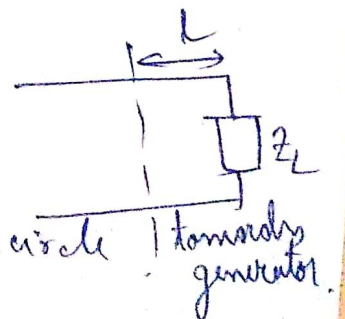
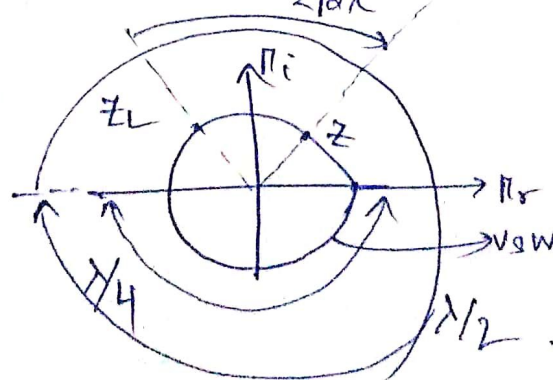
$$(\Gamma'_o, \Gamma'_i) \rightarrow \Gamma_L$$

$$\Gamma_L \rightarrow \Gamma'_L \text{ \& } \theta'_L$$

4) For a given Z_L from load

$$\bar{Z}_L = \frac{Z_L}{Z_0}$$

calculate $\bar{Z}'_L$ at a distance l



l = something
calculate $2\beta l$

Intersection pt on VSWR $\rightarrow Z$.

Characteristic Admittance :- $Y_0 = \frac{1}{Z_0}$

$$\bar{Y} = \frac{Y}{Y_0} = \text{normalised admittance}$$

$$\bar{Y}_L = \frac{Y_L}{Y_0} = \text{" load "}$$

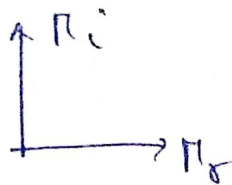
$$\Gamma = \frac{Z - Z_0}{Z + Z_0} = \frac{\frac{1}{Y} - \frac{1}{Y_0}}{\frac{1}{Y} + \frac{1}{Y_0}} = \frac{Y_0 - Y}{Y_0 + Y}$$

$$= \frac{1 - \bar{Y}}{1 + \bar{Y}} = - \left[\frac{\bar{Y} - 1}{\bar{Y} + 1} \right] \quad \text{for any admittance}$$

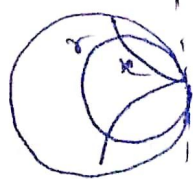
∴ If we rotate 180° on Smith chart (over VSWR circle) then we get normalised admittance for every impedance

∴ For Normalised impedance | For normalised admittance

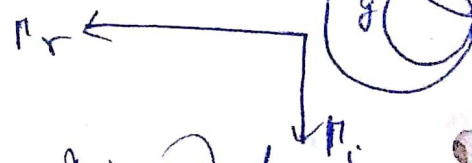
$$\bar{Z} = r + jx$$



r: resistance
x: reactance

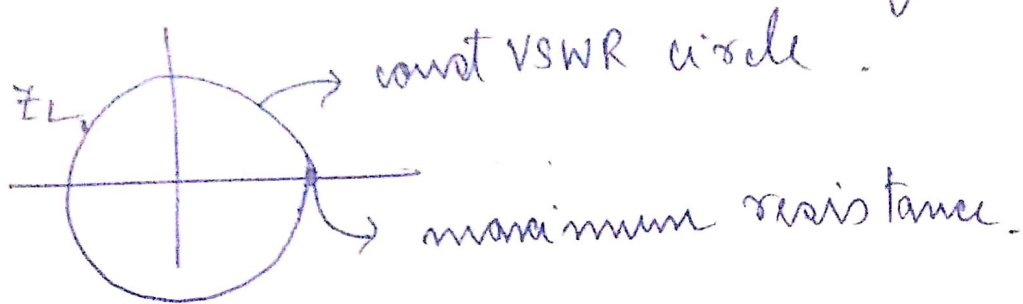


$$\bar{Y} = g + jb$$



g: conductance
b: susceptance

We know, $VSWR = \frac{R_L}{Z_0}$ or $S = \frac{Z_L}{Z_0} = \frac{R_L}{Z_0} = \bar{R}_L$
 when $Z_L > Z_0$ i.e. Z_L is highest impedance.

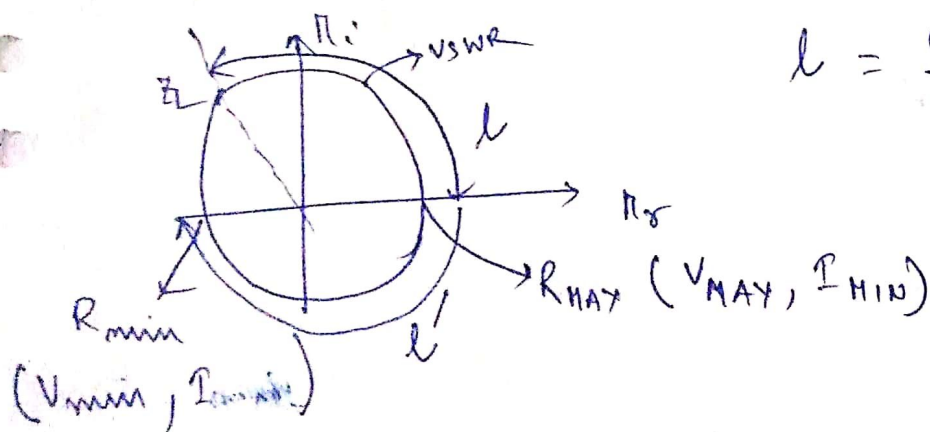


$$S = \frac{Z_0}{Z_{L(\min)}} = \frac{Z_0}{R_{\min}} = \frac{1}{R_{\min}}$$

$$\boxed{R_{\min} = \frac{1}{S}}$$

The leftmost pt on VSWR circle is $Z_{\min}$

- ⑦ distance of voltage & current maximum or minimum ~~on~~ the T_{R_0} line.



$$l = \frac{\text{arc length}}{2\beta}$$

$$l' = \frac{\text{arc length}}{2\beta}$$

$l' \rightarrow$ for $V_{\min}$ $l =$ for $V_{\max}$.

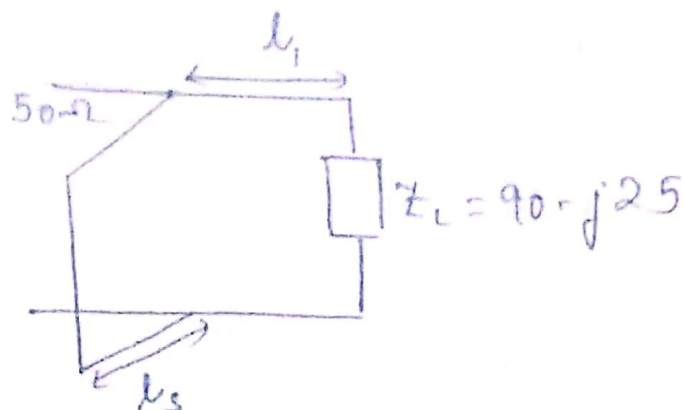
STVB: -

Single stub matching: -

$$Z_L = 90 - j25$$

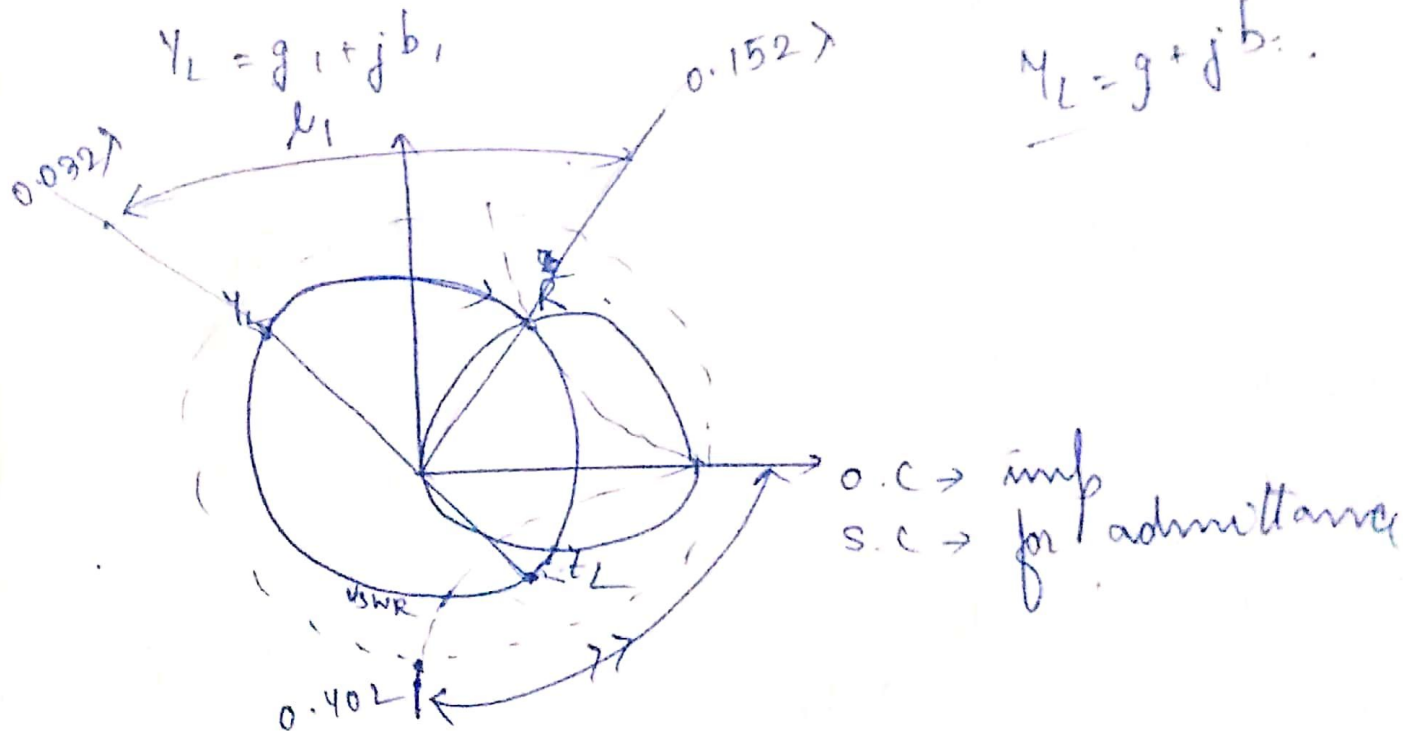
$$\bar{Z}_L = \frac{Z_L}{Z_0} = \frac{90 - j25}{50}$$

$$= 1.8 - j0.5$$



$$Y_L = g + jb$$

$$Y_L = g + jb$$

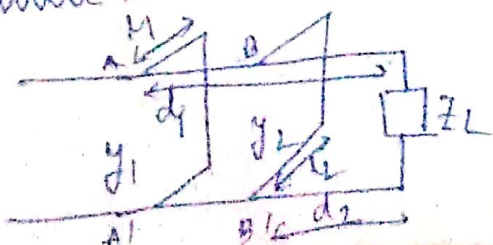


$$R \rightarrow 1 + j0$$

$$l_1 = (0.152 - 0.032 \lambda) = 0.12 \lambda \quad \text{from load}$$

$$l_s = 0.402 - 0.25 = 0.152 \lambda$$

Double stub: -



if load changes no prob one length is fixed & other varying w.r.t Z_L

6.13. SMITH CHARTS

Smith chart is a graphical method to solve the transmission line problems. This chart consists of constant resistance and constant reactance circles. The impedance and admittance which are plotting will be normalised. Let ' Z_L ' is the normalized impedance (note the normalized impedance is shown by lower case letter).

$$\text{Normalized impedance } z_l = \frac{\text{Load impedance}}{\text{Characteristic impedance}}$$

$$= \frac{Z_L}{Z_o} = r + jx \quad \dots (6.90)$$

We know that

$$\rho = \frac{Z_L Z_o}{Z_L + Z_o} \quad \dots (6.91)$$

or

$$\rho = \frac{\frac{Z_L}{Z_o} - 1}{\frac{Z_L}{Z_o} + 1} = \frac{z_l - 1}{z_l + 1}$$

or,

$$\rho = (z_l + 1) = z_l - 1$$

$$\therefore z_l = \frac{1 + \rho}{1 - \rho} \quad \dots(6.92)$$

In polar form 'ρ' can be written as

$$\rho = |\rho| \angle \theta = \rho_r + j \rho_i \quad \dots(6.93)$$

From equation (6.92) and (6.93)

$$z_l = \frac{1 + \rho_r + j \rho_i}{1 - \rho_r - j \rho_i} \quad \dots(6.94)$$

From equation (6.93) and (6.94)

$$\begin{aligned} r + jx &= \frac{(1 + \rho_r) + j \rho_i}{(1 - \rho_r) - j \rho_i} \\ &= \frac{(1 + \rho_r) + j \rho_i}{(1 - \rho_r) - j \rho_i} \cdot \frac{(1 - \rho_r) + j \rho_i}{(1 - \rho_r) + j \rho_i} \end{aligned}$$

$$\begin{aligned} r + jx &= \frac{(1 + \rho_r) + (1 - \rho_r) + j \rho_i (1 + \rho_r) + j \rho_i (1 - \rho_r) - \rho_i^2}{(1 - \rho_r)^2 + \rho_i^2} \\ &= \frac{1 - \rho_r^2 + j \rho_i + j \rho_i \rho_r + j \rho_i - j \rho_i \rho_r - \rho_i^2}{(1 - \rho_r)^2 + \rho_i^2} \end{aligned}$$

$$r + jx = \frac{1 - \rho_r^2 - \rho_i^2}{(1 - \rho_r)^2 + \rho_i^2} + j \frac{2 \rho_i}{(1 - \rho_r)^2 + \rho_i^2}$$

Compare real and imaginary terms

$$\therefore r = \frac{1 - \rho_r^2 - \rho_i^2}{(1 - \rho_r)^2 + \rho_i^2} \quad \dots(6.95)$$

$$x = \frac{2 \rho_i}{(1 - \rho_r)^2 + \rho_i^2} \quad \dots(6.96)$$

Now rearrange equation (6.96)

$$r (1 - \rho_r)^2 + \rho_i^2 = 1 - \rho_r^2 - \rho_i^2$$

$$r + r \rho_r^2 - 2 r \rho_r + r \rho_i^2 = 1 - \rho_r^2 - \rho_i^2$$

$$r + r \rho_r^2 + \rho_r^2 - 2 r \rho_r + r \rho_i^2 + \rho_i^2 = 1$$

$$r + (1 + r) \rho_r^2 - 2 r \rho_r + \rho_i^2 (1 + r) = 1$$

$$\text{or, } \frac{r}{(1 + r)} + \rho_r^2 - \frac{2r}{(1 + r)} \rho_r + \rho_i^2 = \frac{1}{1 + r}$$

Subtract $\frac{r}{(1 + r)^2}$ on both sides

$$\frac{r}{1+r} - \frac{r}{(1+r)^2} + \rho_r^2 - \frac{2r}{(1+r)} \rho_r + \rho_i^2 = \frac{r}{1+r} - \frac{r}{(1+r)^2}$$

$$\frac{r^2}{(1+r)^2} + \rho_r^2 - \frac{2r}{(1+r)} \rho_r + \rho_i^2 = \frac{1}{(1+r)^2}$$

$$\left(\rho_r - \frac{r}{1+r} \right)^2 + \rho_i^2 = \left(\frac{1}{1+r} \right)^2$$

Rearrange equation (6.97)

$$x(1 + \rho_r^2 - 2\rho_r) + x \rho_i^2 = 2\rho_i$$

$$x + x \rho_r^2 - 2x \rho_r + x \rho_i^2 = 2\rho_i$$

Divide this equation by x , we get

$$1 + \rho_r^2 - 2\rho_r + \rho_i^2 = \frac{2}{x} \rho_i$$

$$1 + \rho_r^2 - 2\rho_r + \rho_i^2 - \frac{2}{x} \rho_i = 0$$

On both sides add $\frac{1}{x^2}$

$$(1 + \rho_r^2 - 2\rho_r) + \left(\rho_i^2 - \frac{2}{x} \rho_i + \frac{1}{x^2} \right) = \frac{1}{x^2}$$

$$(\rho_r - 1)^2 + \left(\rho_i - \frac{1}{x} \right)^2 = \frac{1}{x^2}$$

...(6.98)

Equation (6.97) and (6.98) are the equation of the circle.
For constant resistance circle

Centre at $(\rho_r, \rho_i) = \left(\frac{r}{1+r}, 0 \right)$

$$\text{Radius} = \frac{1}{1+r}$$

Similarly for constant reactance circle

Centre at $\left(1, \frac{1}{x} \right)$

$$\text{Radius} = \frac{1}{x}$$

Figure 6.4 shows the circles for different values of ' r '

Similarly, x -circles shown in Fig. 6.5 for different values of ' x '.

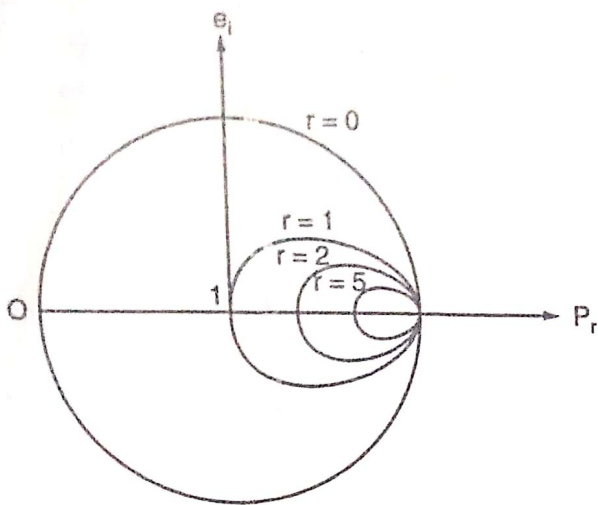


Fig. 6.4.

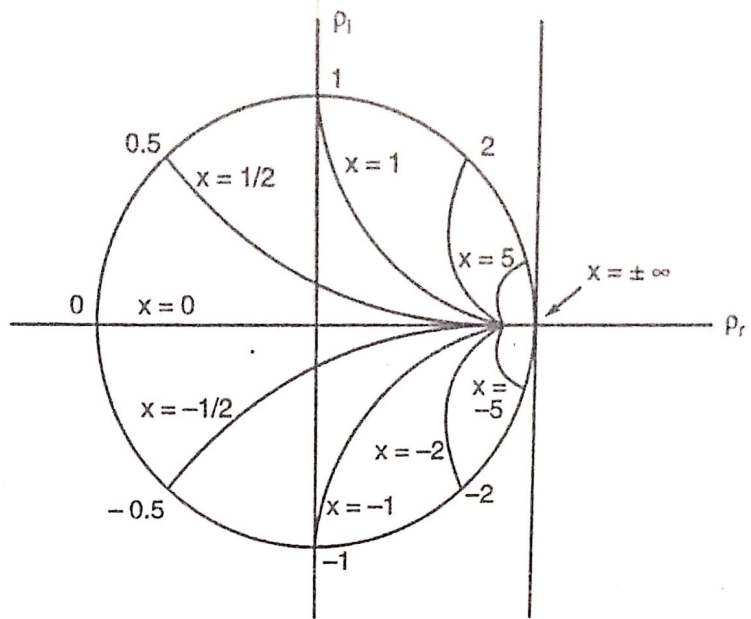


Fig. 6.5.

6.14. CHARACTERISTICS OF SMITH CHART

1. The constant ' r ' and ' x ' loci form two families of orthogonal circles in the chart.
2. The upper half represents $(+jx)$ and lower half represent $(-jx)$.
3. At point 'A' (shown in Fig.) $r = 0$, $x = 0$, i.e., $Z_L = 0$, i.e., point 'A' represents short circuit on the transmission line.
4. At point 'B', $r = \infty$, $x = \infty$, i.e., $Z_L = \infty$, that is point 'B' represents open circuit on the transmission line.
5. A complete revolution around the smith chart represents a distance $\frac{\lambda}{2}$.
6. Clockwise movement on the chart represents the movement toward the generator and counterclockwise movement represents toward the load.
7. The outermost circle on the chart is $R = 0$.
8. As the normalized admittance is reciprocal of the normalized impedance, the corresponding quantities in the admittance chart are 180° out of phase with those in the impedance chart.
9. SWR circle is the circle with centre as the centre of the chart and radius of the circle is the distance between the centre of the chart and the normalized impedance.
10. The line passing through the centre of the chart and given normalized impedance is the load line.
11. The point of intersection of SWR circle and the axis of the chart gives the SWR of the design.
12. $V_{\max}$ occurs on the positive P_r axis, i.e., on OB.
13. $V_{\min}$ occurs on the negative P_r axis, i.e., on OA.
14. By using smith chart we can determine SWR, reflection coefficient, input impedance, input admittance, locations of $V_{\max}$ and $V_{\min}$ and length of the line.

Let us consider an example (using Smith chart).

Q. Find VSWR, (a) $Z_0 = 40 \Omega$

$$Z_L = 70 \Omega$$

(b) $Z_0 = 50 \Omega$

$$Z_L = 30 \Omega$$

Ans)
$$SWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.27$$

$$\therefore VSWR = 1.79$$

$$\Gamma_L' = \frac{Z_L' - Z_0'}{Z_L' + Z_0'} = 0.230$$

$$VSWR = 1.59$$

Q. $SWR = 4$. Find Γ_L .

Ans) $S = \frac{1 + \Gamma_L}{1 - \Gamma_L}$ or $\Gamma_L = \frac{S - 1}{1 + S} = 0.6$

Q. At 1200 cycles Z_0 & γ are $Z_0 = 6.50 - j150$.
 $\gamma = 0.005 + j0.0007$ for a open circuited Tx line.
calculate R, L, C, G .

Ans) $Z_0 = 6.50 - j150$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{Z}{Y}}$$

$$\gamma = 0.005 + j0.007$$

$$(Z_0)(\gamma) = Z = R + j\omega L \Rightarrow$$

$$\frac{\gamma}{Z_0} = Y = G + j\omega C$$

$$\begin{aligned}
 R + j\omega L &= (Z_0)(8) = (650 - j150)(0.0005 + j0.0007) \\
 &= 666 \angle -13 + (8.5 \times 10^{-3}) \angle 45 \\
 &= (0.666)(8.5) \angle 32 = 4.8 + j3
 \end{aligned}$$

$$R = 4.8 \quad \omega L = 3$$

$$\therefore L = \frac{3}{\omega} = \frac{3}{2\pi f} = \frac{3}{2\pi(1200)} = 0.938$$

$$G + j\omega C = (6.75 + j1.8)(10^{-6})$$

$$G = 6.75$$

$$C = \frac{10.8 \times 10^{-16}}{2\pi 1200} = 0.00143 \text{ F/m}$$

A uniform plane wave propagating in a medium has
 $E = 2e^{-\alpha z} \sin(10^8 t - \beta z) \hat{a}_y \text{ V/m}$ ①

If the medium is characterised by $\epsilon_r = 1, \mu_r = 202$
 $\sigma = 3 \text{ S/m}$. calculate α, β & H .

Ans $\frac{\sigma}{\omega \epsilon} = \frac{3}{10^8 \times 1 \times \frac{10^{-9}}{36\pi}} = 3393771$

Hence medium is good conductor.

$$\alpha = \beta = \sqrt{\frac{\mu \omega \sigma}{2}} = \left[\frac{4\pi \times 10^{-7} \times 202 (10^8) (3)}{2} \right]^{1/2}$$

$$= 61.4$$

$$\alpha = 61.4 \text{ Np/m} \quad \beta = 61.4 \text{ rad/m}$$

$$|\eta| = \sqrt{\frac{\mu \omega}{\sigma}} = \left[\frac{4\pi \times 10^{-7} \times 202 (10^8)}{3} \right]^{1/2}$$

$$= \sqrt{\frac{800\pi}{3}}$$

$$\tan 2\theta_m = \frac{\sigma}{\omega \epsilon} = 3393 \rightarrow \theta_m = 45^\circ = \pi/4$$

$$a_H = a_K \times a_E = a_z \times a_y = -a_x$$

$$H_0 = \frac{E_0}{|\eta|} = 2 \sqrt{\frac{3}{800\pi}} = 69.1 \times 10^{-3}$$

$$\therefore H = -69.1 e^{-61.4 z} \sin(10^8 t - 61.4 z - \pi/4) \hat{a}_x \text{ mA/m}$$

Q1) A uniform plane wave in a medium has $\sigma = 10^{-3} \text{ S/m}$, $\epsilon = 80\epsilon_0$, $\mu = \mu_0$ is having a freq of 10 kHz . calculate the diff parameter of wave.

Ans) $\sigma = 10^{-3} \text{ S/km}$ $\epsilon = 80\epsilon_0$ $\mu = \mu_0$

$$\frac{\sigma}{\omega \epsilon} = \frac{10^{-3}}{(2\pi \times 10^4) \times 80 \times 8.854 \times 10^{-12}} = 22.4777$$

Good conductor.

$$\alpha = \beta = \sqrt{\pi f \mu \sigma} = 2\pi \times 10^{-3}$$

$$\eta = \sqrt{\frac{\omega \mu}{\sigma}} \angle 45^\circ = 2\pi(1+j) \Omega$$

$$\lambda = \frac{2\pi}{\beta} = 1000 \text{ m}$$

$$v = \frac{\omega}{\beta} = 10^7 \text{ m/s}$$

Q2) For a non mag material having $\epsilon_r = 2.25$, $\sigma = 10^{-4} \text{ S/m}$. Find (i) loss tangent, (ii) attenuation and (iii) phase const. (iv) η for a wave travelling at freq $f = 25 \text{ MHz}$. Assume the material to be a good dielectric.

Ans) loss tangent, $\frac{\sigma}{\omega \epsilon} = 0.320$

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} = 0.01256 \text{ Np/m}$$

$$\beta = \omega \sqrt{\mu \epsilon} \left(1 + \frac{\sigma^2}{8\omega^2 \epsilon^2} \right) = 0.0796 \text{ rad/m}$$

$$= \sqrt{\mu/\epsilon} \left(1 + j \frac{\sigma}{2\omega\epsilon} \right) = \sqrt{\frac{4\pi \times 10^{-7}}{2.25 \times 8.8 \times 10^{-12}}} \left(1 + j \frac{0.320}{2} \right)$$

$$= 254.35 \angle 9^\circ \Omega$$

Q1) Find the skin depth δ at a freq 1.6 MHz in Al, where $\sigma = 38.2 \text{ Ms/m}$ & $\mu_r = 1$. Also find the prop. const & wave vel

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = 0.06438 \text{ mm}$$

$$\alpha = \beta = \delta^{-1}$$

$$\gamma = 15.53 \times 10^3 + j 15.53 \times 10^3$$

$$= 21.96 \times 10^3 \angle 45^\circ \text{ m}^{-1}$$

$$v = \frac{\omega}{\beta} = \omega \delta = 647.2 \text{ m/s}$$

$$E = E_0 \cos(\omega t - \beta x) a_y$$

$$H = H_0 \cos(\omega t - \beta x) a_z$$

$$= \frac{E_0}{\eta} \cos(\omega t - \beta x) a_z$$

$$P_{\text{re}} = \frac{E_0^2}{\eta} \cos^2(\omega t - \beta x)$$

$$P_{\text{avg}} = \frac{1}{T} \int_0^T \frac{E_0^2}{\eta} \cos^2(\omega t - \beta x) dt \quad T = \frac{1}{f}$$

$$= f \int_0^{1/f} \frac{E_0^2}{\eta} \cos^2(\omega t - \beta x) dt$$

$$= \frac{f}{2} \frac{E_0^2}{\eta} \int_0^{1/f} 2 \cos^2(\omega t - \beta x) dt$$

$$= \frac{1}{2} \frac{E_0^2}{\eta} \int_0^L (1 + \cos(2\omega t - 2\beta x)) dx$$

$$= \frac{1}{2} \frac{E_0^2}{\eta} \left[x + \frac{1}{2\omega} \sin(2\omega t - 2\beta x) \right]_0^L$$

$$= \frac{E_0^2 L}{2\eta} \cdot f \cdot \frac{1}{f} + 0 \quad \left(\begin{array}{l} \sin 2\pi = 0 \\ \sin 0 = 0 \end{array} \right)$$

$$\begin{aligned} \omega &= 2\pi f \\ x &= \frac{1}{f} \\ \omega t &= 2\pi \end{aligned}$$

$$\boxed{P_{avg} = \frac{1}{2} \frac{E_0^2}{\eta}} \text{ Wb/m}^2$$

$$P_{avg} = \frac{E_0^2}{2\eta}$$

$$P_{avg} = \frac{E_0^2}{2\eta}$$

Poynting Theorem

(4)

$$E = E_0 e^{-\alpha z} \cos(\omega t - \beta z) a_x$$

$$H = \frac{E_0}{\eta} e^{-\alpha z} \cos(\omega t - \beta z) a_y$$

$$P = \frac{E_0^2}{\eta} e^{-2\alpha z} \cos^2(\omega t - \beta z) a_z$$

$$P_{avg} = \frac{1}{T} \int_0^T P dt = \frac{1}{T} \int_0^T \frac{E_0^2}{\eta} e^{-2\alpha z} \cos^2(\omega t - \beta z) dt$$

$$= \frac{1}{2T} \frac{E_0^2}{\eta} e^{-2\alpha z} \int_0^T 2 \cos^2(\omega t - \beta z) dt a_z$$

$$= \frac{1}{2T} \frac{E_0^2}{\eta} e^{-\alpha z} \left[\int_0^T (1 + \cos 2(mt - \beta z)) dt \right]$$

$$= \frac{1}{2T} \frac{E_0^2}{\eta} e^{-\alpha z} \left[t \right]_0^T + \frac{1}{2T} \frac{E_0^2}{2\eta} e^{-\alpha z}$$

$$\left[\frac{\sin 2(mt - \beta z)}{2\omega} \right]_0^T$$

$$T = \frac{2\pi}{\omega}$$

$$\text{or } \omega = \frac{2\pi}{T}$$

$$\text{and } \beta = \frac{\omega}{u} = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^6}$$

$$= 60 \text{ m}$$

$$P_{\text{avg}} = \frac{1}{2T} \frac{E_0^2}{\eta} e^{-2\alpha z} T$$

$$= \frac{E_0^2}{\eta} e^{-2\alpha z}$$

$$P_{\text{avg}}(z) = \frac{E_0^2}{|\eta|} e^{-2\alpha z} \text{ Wb/m}^2$$

AVERAGE POWER ↑

$$= \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \cos \theta \sin \theta$$

Q6 a) Write short note on

- (i) Smith chart (ii) Quarter Wave Transformer (8)
b) Write Boundary conditions for electric & Magnetic field in mathematical form and also write the related statements. (7)

Answer (i) SMITH CHART : —

Smith chart is a graphical method to solve the transmission line problems. The chart consists of constant resistance & constant reactance circles. All impedances and admittances are to be normalised to characteristic impedances & characteristic admittances resp. ($\bar{Z}_L = \frac{Z_L}{Z_0}$ & $\bar{Y}_L = \frac{Y_L}{Y_0}$) — (1)

We know $Z_L = r + jx$ (r: resistance, x: reactance)

$$\& \Gamma = \Gamma_r + j\Gamma_i \text{ (reflection-co-efficient)}$$

All impedances are transformed from complex impedance plane to complex reflection co-efficient plane (Γ plane)

$$\Gamma = \Gamma_r + j\Gamma_i = |\Gamma| \angle \theta_\Gamma \quad \text{--- (2)}$$

$$\text{We know, } \bar{Z}_L = \frac{1 + \Gamma}{1 - \Gamma} \quad \text{--- (3)}$$

$$r + jx = \frac{1 + \Gamma_r + j\Gamma_i}{1 - (\Gamma_r + j\Gamma_i)} = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}$$

$$= \frac{[(1 + \Gamma_r) + j\Gamma_i][(1 - \Gamma_r) + j\Gamma_i]}{(1 - \Gamma_r)^2 - (j\Gamma_i)^2}$$

$$r + jx = \frac{1 - \Gamma_r^2 - \Gamma_i^2 + 2j\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2} = \frac{(1 - \Gamma_r^2) - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} + j \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}$$

Comparing L.H.S with R.H.S,

$$r = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2} \quad \& \quad x = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2} \quad \text{--- (4)}$$

$$\text{--- (5)}$$

Solving (4) we get

$$\left(\Gamma_r - \frac{r}{r+1}\right)^2 + \Gamma_i^2 = \left(\frac{1}{r+1}\right)^2 \quad \text{--- (6)}$$

Solving we get an equation of circle $[(x-h)^2 + (y-k)^2 = a^2]$ with centre at $\left(\frac{r}{r+1}, 0\right)$ & radius at $\left(\frac{1}{r+1}\right)$.

Similarly solving equation (5) we get

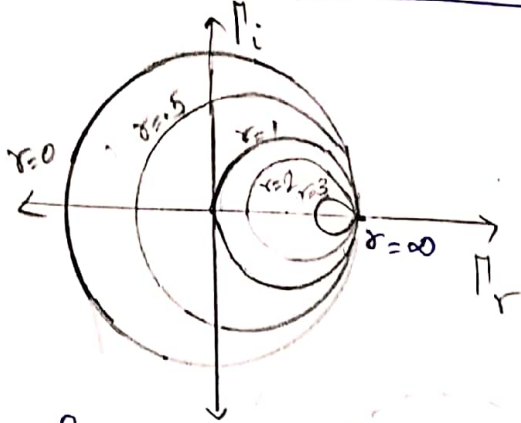
$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x}\right)^2 = \left(\frac{1}{x}\right)^2 \quad \text{--- (7)}$$

eq (7) is equation of reactance circle with centre at 1 & radius $\left(\frac{1}{x}\right)$.

In any transmission line problem $|\Gamma| = 1$. Hence in complex Γ plane first a unit circle (radius = 1) is drawn. All points has to be plotted in unit circle.

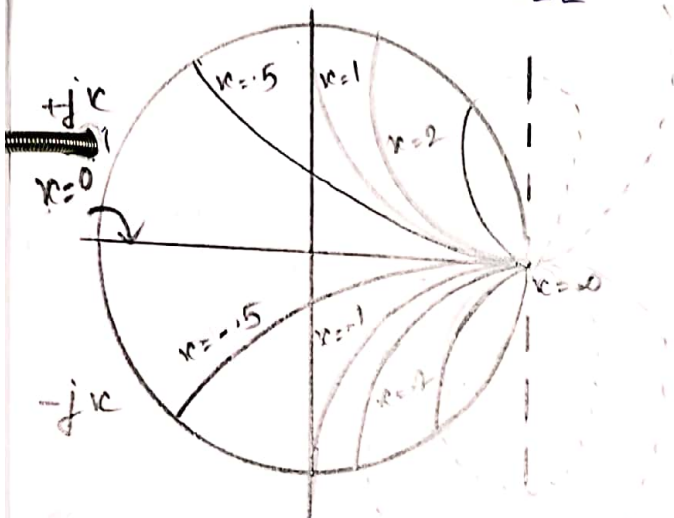
&

COMPLEX Γ -PLANE:—

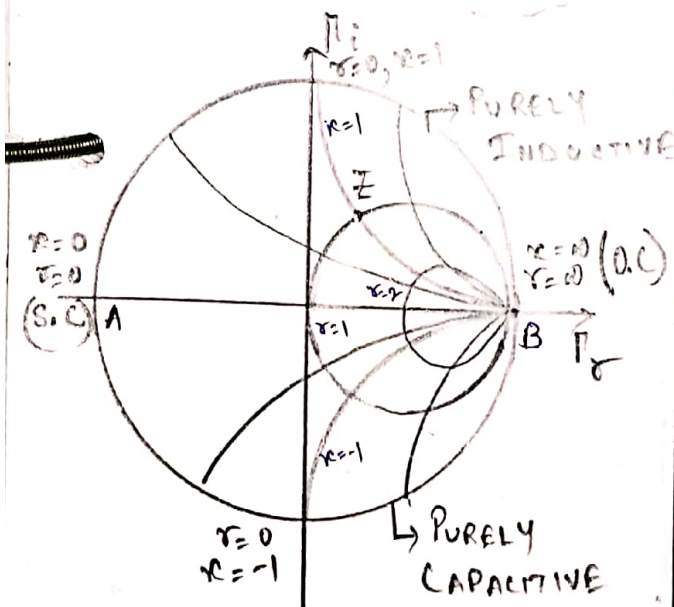


RESISTANCE CIRCLE

r	RADIUS ($\frac{1}{r+1}$)	CENTRE ($\frac{r}{r+1}, 0$)
0	1	0, 0
1	0.5	0.5, 0
2	$\frac{1}{3}$	$\frac{2}{3}, 0$
$\vdots$	$\vdots$	$\vdots$
∞	$\frac{1}{\infty+1} = 0$ (pt)	1, 0



x	RADIUS ($\frac{1}{x}$)	CENTRE ($1, \frac{1}{x}$)
0	∞	1, $\infty \Rightarrow$ line
1	1	1, 1
2	$\frac{1}{2}$	1, $\frac{1}{2}$
$\vdots$	$\vdots$	$\vdots$
∞	0 (pt)	1, 0
-1	-1	1, -1



$$\begin{aligned} \bar{Z}_L &= r + jx \\ \bar{Y}_L &= g + jb \end{aligned}$$

& scales at the periphery to measure distance.

1) A complete revolution around Smith chart represents a distance of $\frac{1}{2}$

2) For Impedance A is short circuited & B is open-circuited & for admittance A is o.c & B is S.C.

3) Any $\bar{Z}_L$ & $\bar{Y}_L$ can be mapped according to corresponding values (r, x) & (g, b)

4) clockwise movement $\rightarrow$ movement towards generator

5) anticlockwise $\rightarrow$ movement towards load.